

Generation Interconnection System Impact Study Report

For Queue Project Q15-0518

Clover 345 kV
March 03, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0518. The Interconnection Customer proposes a 75.0 MW solar photovoltaic facility interconnecting at the Clover 345 kV substation in the Virginia area. The request was received on September 12, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 249 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,184,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GLENANNAN - CAL SIERRA DEVELOPMENT 345 KV L	ROBERTS SW STA - GLENANNAN 345 KV	113.0	1,319	8.0
MCCARTNEY - GASTON 345 KV	MCCARTNEY SUBSTATION - ROANOKE RAPIDS 345 KV	109.1	1,322	7.5
CARRABELLE SUBSTATION - LOUISA GENERATION FACILITY 345 KV	LOUISA GENERATION FACILITY - NORTH ANNA 345 KV line	107.1	1,907	11.5
ROBERTS SW STA - NAUVOO 345 KV	NAUVOO - CITY W DIESEL PLANT 345 KV	106.2	1,505	8.5
BIXBY HILL - TOMBSTONE 500 KV	TOMBSTONE - SHARYLAND 345 KV	105.6	3,937	8.4
ROBERTS SW STA - GLENANNAN 345 KV	CAL SIERRA DEVELOPMENT SUBSTATION - GLENANNAN 345 KV	104.2	1,461	8.6
SHINGLETOWN - RIVA RD 345 KV	TABER - GLENANNAN 345 KV	104.2	1,628	6.9
BOONSBORO - NACOOCHEE 138 KV	BEAR - CLOVER 345 KV	101.9	379	5.7
RUSIC - HUGHSON 345 KV	RUSTIC - DIXLETA 345 KV Line	101.6	1,970	12.5
SOUTH TEXA - NACOOCHEE SUBSTATION 138 KV	CLOVER - BEAR 345 KV	101.5	382	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GLENANNAN - CAL SIERRA DEVELOPMENT 345 KV	Rebuild line with higher-capacity conductor	\$63,946,000	1% (\$490,000)
MCCARTNEY - GASTON 345 KV	Reconductor line	\$29,245,000	1% (\$172,000)
CARRABELLE SUBSTATION - LOUISA GENERATION FACILITY 345 KV	Reconductor line	\$31,748,000	1% (\$257,000)
NAUVOO SUBSTATION - ROBERTS SW STA 345 KV	Reconductor line	\$13,539,000	1% (\$111,000)
TOMBSTONE - BIXBY HILL 500 KV	Reconductor line	\$68,107,000	1% (\$513,000)
ROBERTS SW STA - GLENANNAN 345 KV	Reconductor line	\$19,410,000	1% (\$150,000)
SHINGLETOWN - RIVA ROAD 345 KV line	Reconductor line	\$29,924,000	1% (\$234,000)
BOONSBORO SUBSTATION - NACOOCHEE 138 KV Line	Reconductor line	\$1,812,000	3% (\$61,000)
RUSIC - HUGHSON 345 KV	Reconductor line	\$16,951,000	1% (\$129,000)
SOUTH TEXA - NACOOCHEE SUBSTATION 138 KV	Reconductor line	\$1,987,000	3% (\$67,000)
	Total allocated to project		\$2,184,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.